

Sampling methods for official surveys

Trainer note for Statistical Investigators. Prepared as sample material for the Kaushal AI prototype. Not an official MoSPI publication.

1. Population, frame and unit

The target population is the complete set of units about which a survey wants to draw conclusions. A sampling frame is the list or map from which the sample is actually selected. When the frame leaves out part of the target population, the survey suffers from undercoverage, and no increase in sample size can correct it. The sampling unit is the unit selected at each stage, such as a village, a block or a household.

2. Probability sampling

In probability sampling every unit in the frame has a known, non-zero chance of selection. This property allows the survey to produce unbiased estimates and to measure their sampling error. Non-probability methods such as convenience or quota sampling cannot support a valid estimate of sampling error.

In simple random sampling without replacement, every possible sample of size n has the same chance of being selected. Systematic sampling selects every k -th unit after a random start, where k is the sampling interval obtained by dividing the frame size by the sample size.

3. Stratification

Stratified sampling divides the population into non-overlapping groups called strata and selects a separate sample from each stratum. Stratification improves precision when units within a stratum are similar to each other. Under proportional allocation, the sample size in each stratum is proportional to the stratum population size. Neyman allocation gives larger samples to strata that are larger and more variable.

4. Cluster and multi-stage sampling

Cluster sampling selects groups of units, such as villages or urban blocks, instead of individual units. It reduces travel and listing cost, but units in the same cluster tend to be alike, so a cluster sample is usually less precise than a simple random sample of the same size. Large national household surveys commonly

use multi-stage designs, where first-stage units such as villages are selected first and households are then selected within them.

Selection with probability proportional to size, often written as PPS, gives larger first-stage units a higher chance of selection. The size measure is usually a population count from the most recent census or listing.

5. Design effect and weights

The design effect is the ratio of the variance of an estimate under the actual design to the variance under a simple random sample of the same size. A design effect of 2 means the design needs twice as many units to reach the precision of a simple random sample.

The design weight of a unit is the inverse of its probability of selection. A household selected with probability 1 in 500 represents 500 households in the population. Weights are then adjusted for non-response so that responding units also represent the units that did not respond.

6. Sampling and non-sampling error

Sampling error arises because only a part of the population is observed, and it falls as the sample size grows. Non-sampling errors include coverage error, non-response, measurement error and processing error. Non-sampling errors do not necessarily fall as the sample grows, so field supervision, clear questionnaires and scrutiny of data remain essential.

The standard error measures how much an estimate would vary across repeated samples drawn with the same design. The relative standard error, also called the coefficient of variation of the estimate, is the standard error divided by the estimate, and is widely used to judge whether an estimate is reliable enough to publish.